

Diophantine Certificates for Section 6

Exact witnesses for the element endpoint

This document records exact integral solutions for every parameter triple in the computational proposition of Section 6 except

$$(r, p_1, p_2) = (13, 11, 61) \quad \text{and} \quad (37, 67, 97).$$

These two exceptional triples are handled separately by the programs

$$\text{check_11_61.txt} \quad \text{and} \quad \text{check_67_97.txt},$$

which recover generators with `bnfisprincipal` and verify the corresponding norm identities exactly.

All equalities below are identities in integers and are therefore unconditional. They show unconditionally that the element endpoint is at most the listed exponent r . As explained in Section 6 of the manuscript, identifying the same r as the first principal-ideal endpoint, and hence concluding that the two endpoints are equal, remains conditional on GRH.

1. Case (I): $p_1 \equiv 3 \pmod{8}$ and $p_2 \equiv 5 \pmod{8}$

The relevant equation is

$$p_2 y^2 + p_1 z^2 = 2^{r+2}.$$

For an integral solution (y, z) , the element

$$x = \frac{z\sqrt{-p_1} + y\sqrt{p_2}}{2}$$

belongs to the relevant biquadratic intermediate field and satisfies

$$x\bar{x} = \frac{p_1 z^2 + p_2 y^2}{4} = 2^r.$$

(a) For $(r, p_1, p_2) = (3, 19, 13)$, take $(y, z) = (1, 1)$. Then

$$13 \cdot 1^2 + 19 \cdot 1^2 = 32 = 2^5 = 2^{3+2}.$$

Hence

$$x = \frac{\sqrt{-19} + \sqrt{13}}{2}, \quad x\bar{x} = 8 = 2^3.$$

(b) For $(r, p_1, p_2) = (11, 67, 13)$, take $(y, z) = (25, 1)$. Then

$$13 \cdot 25^2 + 67 \cdot 1^2 = 8192 = 2^{13} = 2^{11+2}.$$

Hence

$$x = \frac{\sqrt{-67} + 25\sqrt{13}}{2}, \quad x\bar{x} = 2048 = 2^{11}.$$

2. Quadratic certificates for $p_1 \equiv 7 \pmod{8}$

For all five parameters in this section, the relevant quadratic equation is

$$u^2 + p_1 z^2 = 2^{r+2},$$

and one may take $z = 1$. The element

$$x = \frac{u + z\sqrt{-p_1}}{2}$$

belongs to the relevant imaginary quadratic intermediate field and satisfies

$$x\bar{x} = \frac{u^2 + p_1 z^2}{4} = 2^r.$$

2.1. Case (2) of Feng's Theorem 4.1: $p_2 \equiv 5 \pmod{8}$

(a) For $(r, p_1, p_2) = (7, 71, 61)$, take $(u, z) = (21, 1)$. Then

$$21^2 + 71 \cdot 1^2 = 512 = 2^9 = 2^{7+2}.$$

(b) For $(r, p_1, p_2) = (5, 79, 37)$, take $(u, z) = (7, 1)$. Then

$$7^2 + 79 \cdot 1^2 = 128 = 2^7 = 2^{5+2}.$$

(c) For $(r, p_1, p_2) = (5, 79, 61)$, take $(u, z) = (7, 1)$. Again,

$$7^2 + 79 \cdot 1^2 = 128 = 2^7 = 2^{5+2}.$$

2.2. Case (II) of the current endpoint-extension theorem: $p_2 \equiv 1 \pmod{8}$

(a) For $(r, p_1, p_2) = (3, 31, 17)$, take $(u, z) = (1, 1)$. Then

$$1^2 + 31 \cdot 1^2 = 32 = 2^5 = 2^{3+2}.$$

(b) For $(r, p_1, p_2) = (7, 71, 41)$, take $(u, z) = (21, 1)$. Then

$$21^2 + 71 \cdot 1^2 = 512 = 2^9 = 2^{7+2}.$$

3. Case (III): $p_1 \equiv 3 \pmod{8}$ and $p_2 \equiv 1 \pmod{8}$

Write $p_2 = A^2 + B^2$ with $A \equiv 1 \pmod{4}$. The quartic norm-form criterion in Appendix A of the manuscript is the system

$$2^{r+4} = p_1^2 X^2 + p_2 Y^2 + 2p_1 p_2 (Z^2 + W^2),$$

$$XY + 2BZW + A(W^2 - Z^2) = 0.$$

An integral solution (X, Y, Z, W) is equivalent, by that norm-form calculation, to an element x in the relevant quartic intermediate field such that $x\bar{x} = 2^r$.

(a) For $(r, p_1, p_2) = (23, 19, 97)$, take $A = 9$ and $B = 4$, so that $97 = 9^2 + 4^2$, and take

$$(X, Y, Z, W) = (303, 3, -139, -90).$$

The first equation becomes

$$\begin{aligned} 19^2 \cdot 303^2 + 97 \cdot 3^2 + 2 \cdot 19 \cdot 97((-139)^2 + (-90)^2) \\ = 134217728 = 2^{27} = 2^{23+4}, \end{aligned}$$

and the second equation becomes

$$303 \cdot 3 + 2 \cdot 4 \cdot (-139) \cdot (-90) + 9((-90)^2 - (-139)^2) = 0.$$

(b) For $(r, p_1, p_2) = (7, 11, 41)$, take $A = 5$ and $B = 4$, so that $41 = 5^2 + 4^2$, and take

$$(X, Y, Z, W) = (1, 5, -1, 0).$$

The first equation becomes

$$11^2 \cdot 1^2 + 41 \cdot 5^2 + 2 \cdot 11 \cdot 41((-1)^2 + 0^2) = 2048 = 2^{11} = 2^{7+4},$$

and the second equation becomes

$$1 \cdot 5 + 2 \cdot 4 \cdot (-1) \cdot 0 + 5(0^2 - (-1)^2) = 0.$$

4. Conclusion

The nine nonexceptional triples in the computational proposition of Section 6 therefore admit explicit integral solutions to the corresponding quadratic, biquadratic, or quartic norm-form equations. These certificates yield elements x satisfying $x\bar{x} = 2^r$ in the indicated intermediate fields, hence also in the full descent fields used in Section 6.